

Employee ID	Department	Designation	Date of Joining	Salary				
E001	HR	AVP	1/1/2017	\$40,000				
E002	Analytics	AVP	12/1/2016	\$75,000				
E003	Sales	AVP	12/1/2016	\$30,000				
E004	Analytics	AVP	11/1/2016	\$80,000				
E005	HR	Analyst	10/1/2016	\$35,000				
E006	Sales	Analyst	11/1/2016	\$25,000				
E007	Analytics	Analyst	10/1/2016	\$60,000				
E008	Sales	Analyst	11/1/2016	\$20,000				
E009	IT	Analyst	10/1/2016	\$40,000				
E010	IT	Analyst	10/1/2016	\$38,000				
			Result	Formula				
Salary for Analytics Group			\$225,000		225000			=SUMIF(C2:C10,"Analytics",D2:D10)
Exercise 2								
Use SUMIF function to calculate salary of all employees joined on 1st Oct 2016								
Employee ID	Department	Designation	Date of Joining	Salary				
E001	HR	AVP	1/1/2017	\$40,000				
E002	Analytics	AVP	12/1/2016	\$75,000				
E003	Sales	AVP	12/1/2016	\$30,000				
E004	Analytics	AVP	11/1/2016	\$80,000				
E005	HR	Analyst	10/1/2016	\$35,000				
E006	Sales	Analyst	11/1/2016	\$25,000				
E007	Analytics	Analyst	10/1/2016	\$60,000				
E008	Sales	Analyst	11/1/2016	\$20,000				
E009	IT	Analyst	10/1/2016	\$40,000				
E010	IT	Analyst	10/1/2016	\$38,000				
			Result	Formula				
Salary for Analytics Group			\$173,000		173000			=SUMIF(D2:D10,">=10/1/2016",E2:E10)

Exercise 1									
Use SUM function to calculate sum of the following values									
Values to be summed					Result				
	1				801	=SUM(A1:A5,B6:C5)			
	100			500					
		200							
Exercise 2									
Use IF and SUM function to calculate sum of the following values if conditions meet. If the conditions do not meet, then return value 0. First has been done as an example									
Condition to meet	Col B	Col C	Col D	Col E	Result				
Col B > 1	5	1	2	4	12	=IF(B5>1,SUM(B5:E5),0)			
Col C > 50	9	56	6	12	0	=IF(B14>50, SUM(B14:E14),0)			
Col C > 500	12	345	23	547	0	=IF(B15>500, SUM(B15:E15),0)			
Col E > 80	84.3	59.8	80.9	80.5	305.5	=IF(B18>80, SUM(B18:E18),0)			
Col D > 35	9.8	0.2	38	80.9	0	=IF(B17>35, SUM(B17:E17),0)			
Exercise 3									
Use IF, AND and SUM functions to calculate sum of the following values if conditions meet. If the conditions do not meet, then return value 0. First has been done as an example									
First Condition	Second Condition	Col B	Col C	Col D	Col E	Result			
Col B > 1	Col D > 2	5	1	2	4	12	=IF(AND(C6>1,E6>2),SUM(C6:F6),0)		
Col C > 50	Col A > 5	9	56	6	12	0	=IF(AND(C25>50, E25>5), SUM(C25:F25),0)		
Col C > 500		12	345	23	547	0	=IF(AND(C26>500, E26>500), SUM(C26:F26),0)		
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5	305.5	=IF(AND(C27>80, E27>80), SUM(C27:F27),0)		
Col D > 35	Col B > 9.8	9.8	0.2	38	80.9	0	=IF(AND(C28>35, E28>9.8), SUM(C28:F28),0)		
Exercise 4									
Use IF, OR and SUM functions to calculate sum of the following values if either of the 2 conditions meet. If the conditions do not meet, then return value 0. First has been done as an example									
First Condition	Second Condition	Col B	Col C	Col D	Col E	Result			
Col B > 1	Col D > 1	5	1	2	4	12	=IF(OR(C76>1,E76>1),SUM(C76:F76),0)		
Col C > 50	Col A > 5	9	56	6	12	83	=IF(OR(C36>50, E36>5), SUM(C36:F36),0)		
Col C > 500	Col E > 500	12	345	23	547	0	=IF(AND(C37>500, E37>500), SUM(C37:F37),0)		
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5	305.5	=IF(AND(C38>80, E38>80), SUM(C38:F38),0)		

Exercise 1

Below are 3 different tables, in the first table we have employees and their designation, in the second table we have their names and salary and in the 3rd one we have their names and department. Use Vlookup() to fill all these details in the 4th table.

Table 1		Table 2		Table 3	
E Name	Designation	E Name	Salary	E Name	Dept
Ben	AVP	Ben	\$85,000	Ben	Analytics
Ken	Analyst	Ken	\$60,000	Ken	Analytics
Rachel	AVP	Rachel	\$65,000	Rachel	IT
Katie	Analyst	Katie	\$50,000	Katie	HR
Helen	Analyst	Helen	\$45,000	Helen	IT

Table 4			
E Name	Designation	Salary	Dept
Katie	Analyst	\$50,000	HR
Helen	Analyst	\$45,000	IT
Rachel	AVP	\$65,000	IT
Ben	AVP	\$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analyst	\$60,000	Analytics

Exercise 2

Below in the Lookup table, we have cost for different parts for different car makers. Use VLOOKUP() to get costs of items in the second table.

Lookup Table					
Mercedes	GearBox	Engine	Steering	Ignition	CYHead
Mercedes	\$2,500	\$5,000	\$1,250	\$250	\$300
BMW	\$2,000	\$4,999	\$1,302	\$300	\$290
Audi	\$3,000	\$5,200	\$1,100	\$200	\$310
Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs		\$300	\$2,000	\$3,000	\$1,100
				\$250	\$5,000
					\$2,000
					\$4,999

Col A	Results				
31-Jan	50			=MINUTE(A2)	
1/1/17 0:01	1			=MINUTE(A3)	
12:45 AM	45			=MINUTE(A4)	

Dates	Return Value	Results			
1/1/2017	1	1		=WEEKDAY(A2,1)	
1/1/2017	2	7		=WEEKDAY(A3,2)	
1/1/2017	3	6		=WEEKDAY(A4,3)	
Dates	Weekday	New Date			
3/28/2017	3	3/31/2017		=IF(WEEKDAY(A8,1)=3, A8+3, A8)	
3/29/2017	4	3/29/2017	=WEEKDAY(A9,1)	=IF(WEEKDAY(A9,1)=3,	
3/21/2017	3	3/24/2017	=WEEKDAY(A10,1)	=IF(WEEKDAY(A10,1)=3,	
1/1/2016	6	1/1/2016	=WEEKDAY(A11,1)	=IF(WEEKDAY(A11,1)=3,	
1/1/2017	1	1/1/2017	=WEEKDAY(A12,1)	=IF(WEEKDAY(A12,1)=3,	
1/1/2015	5	1/1/2015	=WEEKDAY(A13,1)	=IF(WEEKDAY(A13,1)=3,	
1/1/2014	4	1/1/2014	=WEEKDAY(A14,1)	=IF(WEEKDAY(A14,1)=3,	
1/1/2013	3	1/4/2013	=WEEKDAY(A15,1)	=IF(WEEKDAY(A15,1)=3,	

Date	Results				
1/15/2016	Use this date to calculate dates from Row 30 onwards				
2/15/2016	2/15/2016			=DATE(YEAR(A2),MONTH(A2)+1,DAY(A2))	
3/15/2016	3/15/2016			=DATE(YEAR(A3),MONTH(A3)+1,DAY(A3))	
4/15/2016	4/15/2016			=DATE(YEAR(A4),MONTH(A4)+1,DAY(A4))	
5/15/2016	5/15/2016			=DATE(YEAR(A5),MONTH(A5)+1,DAY(A5))	
6/15/2016	6/15/2016			=DATE(YEAR(A6),MONTH(A6)+1,DAY(A6))	
7/15/2016	7/15/2016			=DATE(YEAR(A7),MONTH(A7)+1,DAY(A7))	
8/20/2017	8/20/2016			=DATE(YEAR(A8),MONTH(A8)+1,DAY(A8)+5)	
Col A	Results from NOW fuction	Results from Year function			
Current date and time	3/26/26 20:51	2026	=NOW()		=YEAR(B13)
Date and Time 365 days ago	3/26/25 20:51	2025	=NOW()-365		=YEAR(B14)

Exercise 1			
Text	Result	Formula	
Phone Number is +(0)1234	is +(0)1234	is +(0)1234	=RIGHT(A4, 11)
This is used for data cleaning	data cleaning	data cleaning	=RIGHT(A5, 13)
Exercise 2			
Text	Result	Formula	
Phone Number is +(0)1234	Number is +(0)1234	Number is +(0)1234	=RIGHT(A10,LEN(A10)-5)
This is used for data cleaning	is used for data cleaning	is used for data cleaning	=RIGHT(A11,LEN(A11)-5)

Exercise 1			
Text	Result	Formula	
Phone Number is +(0)1234	Phone Number is +(0)	Phone Number is +(0)	=LEFT(A4, 20)
This is used for data cleaning	This is used for data cle	This is used for data cle	=LEFT(A5,25)
Exercise 2			
Text	Result	Formula	
Phone Number is +(0)1234	Phone Number is +(0)	Phone Number is +(0)	=LEFT(A10,LEN(A10)-4)
This is used for data cleaning	This is used for data cle	This is used for data cle	=LEFT(A11,LEN(A11)-5)

Exercise 1		
Text	Formula	
Phone Number is +(0)1234	24	=LEN(A4)
This is used for data cleaning	30	=LEN(A5)

Exercise 1			
Text	Formula		
GOOD	good	=LOWER(A4)	
HOW are you?	how are you?	=LOWER(A5)	
Exercise 2			
Col A	Col B	Result	
MATCH	match	TRUE	=EXACT(LOWER(A9), B9)
MATCH	maTCH	FALSE	=EXACT(UPPER(A10), B10)
CORECT	correct	FALSE	=EXACT(UPPER(A11), B11)

